Cambridge Lower Secondary Mathematics Stage 7

Percentages: 30-Question Applied Practice Quiz Answer Key

Cambridge Assessment International Education Alignment

Instructions

This document contains 30 short-answer and essay-style questions covering Fractions, Decimals, Percentages (FDP), Special Percentages, Percentages of Quantities, Financial Contexts, Handling Data, Probability, and Geometry. Space is provided below each question for your working and answers. The complete answer key with step-by-step solutions is provided at the end of this document.

Questions

1. Convert 68% into a fraction in its simplest form.

2. Express $\frac{7}{20}$ as a percentage.

3. Write 0.625 as a percentage.

4. Convert $\frac{11}{25}$ into a decimal and a percentage. Write your answer separated by a comma.

5. Arrange the following values in ascending order (smallest to largest): $0.42, \frac{2}{5}, 45\%$. Write them separated by commas.

6. A recipe requires $\frac{3}{8}$ of a kilogram of flour. What percentage of a kilogram is this?

7. Write 0.6% as a decimal.

8. Express 0.25% as a fraction in its simplest form.

9. Convert 235% into a mixed number in its simplest form.

10. A rare mineral makes up 0.008 of a rock sample's mass. What percentage of the rock is this mineral?

11. A company's profit this year is 180% of last year's profit. If last year's profit was $\$4,500$, calculate this year's profit in dollars.

12. Calculate 45% of 320 grams.

13. Find 17.5% of $\$400$.

14. A library has $1,200$ books. 65% of the books are fiction, and the rest are non-fiction. How many non-fiction books are in the library?

15. Express 750 meters as a percentage of 3 kilometers.

16. What percentage of 2 hours is 18 minutes?

17. Calculate 0.5% of $6,400$.

18. A bicycle is priced at $\$250$. In a weekend sale, there is a 16% discount. What is the new sale price of the bicycle?

19. A restaurant bill is $\$85$ before tax. A 12% service tax is added to the bill. What is the total bill including tax?

20. Liam buys a vintage watch for $\$120$ and later sells it for $\$156$. Calculate his percentage profit.

21. A laptop costs $\$600$. Its price is first increased by 10%, and then the new price is reduced by 10% in a clearance sale. What is the final price of the laptop?

22. Maya deposits $\$500$ into a bank account that pays 4% simple interest per year. How much total money will she have in her account after 3 years?

23. In School A, 36 out of 45 students play a musical instrument. In School B, 48 out of 64 students play a musical instrument. Which school has a higher percentage of students playing an instrument?

24. In a pie chart showing favorite sports of 200 students, the sector for Football represents 38% and Basketball represents 27%. How many students chose other sports?

25. A waffle diagram has 100 squares. 28 squares are shaded blue, 17 are shaded green, and the rest are unshaded. What percentage of the diagram is unshaded?

26. In a survey of 250 people, 64% said they prefer tea over coffee. How many more people prefer tea than coffee?

27. A spinner has 8 equal sections numbered 1 to 8. What is the percentage probability of spinning a multiple of 4?

28. A bag contains 14 red counters, 11 blue counters, and 25 yellow counters. One counter is picked at random. What is the percentage probability that it is NOT blue?

29. A rectangle measures 12 cm by 5 cm. A square of side 3 cm is cut out from inside the rectangle. What percentage of the rectangle's original area was cut out?

30. A square has a side length of 10 cm. If each side length is increased by 20%, what is the percentage increase in the AREA of the square?

Answer Key & Step-by-Step Solutions

1. **17/25**
   Working: $\frac{68}{100} = \frac{68 \div 4}{100 \div 4} = \frac{17}{25}$.

2. **35%**
   Working: $\frac{7 \times 5}{20 \times 5} = \frac{35}{100} = 35\%$.

3. **62.5%**
   Working: $0.625 \times 100 = 62.5\%$.

4. **0.44, 44%**
   Working: $\frac{11}{25} = \frac{44}{100} = 44\% = 0.44$.

5. **2/5, 0.42, 45%**
   Working: Converting all to percentages gives $40\%, 42\%, 45\%$. Ordering from smallest to largest gives $\frac{2}{5}, 0.42, 45\%$.

6. **37.5%**
   Working: $\frac{1}{8} = 12.5\%$, so $\frac{3}{8} = 3 \times 12.5\% = 37.5\%$.

7. **0.006**
   Working: $0.6 \div 100 = 0.006$.

8. **1/400**
   Working: $\frac{0.25}{100} = \frac{0.25 \times 4}{100 \times 4} = \frac{1}{400}$.

9. **2 7/20**
   Working: $235\% = 2\frac{35}{100} = 2\frac{7}{20}$.

10. **0.8%**
    Working: $0.008 \times 100 = 0.8\%$.

11. **$8100**
    Working: $100\% = \$4,500$, $80\% = 8 \times \$450 = \$3,600$. Total = $\$4,500 + \$3,600 = \$8,100$.

12. **144 g**
    Working: 40% of $320 = 128$, 5% of $320 = 16$. Total = $128 + 16 = 144$ g.

13. **$70**
    Working: $10\% = \$40$, $5\% = \$20$, $2.5\% = \$10$. Total = $\$40 + \$20 + \$10 = \$70$.

14. **420**
    Working: Non-fiction percentage = $100\% - 65\% = 35\%$. 35% of $1,200 = 420$.

15. **25%**
    Working: 3 km $= 3,000$ m. $\frac{750}{3000} = \frac{1}{4} = 25\%$.

16. **15%**
    Working: 2 hours $= 120$ minutes. $\frac{18}{120} \times 100\% = \frac{3}{20} \times 100\% = 15\%$.

17. **32**
    Working: 1% of $6,400 = 64$. Half of $64 = 32$.

18. **$210**
    Working: Discount = 16% of $\$250 = \$40$. Sale price = $\$250 - \$40 = \$210$.

19. **$95.20**
    Working: Tax = 12% of $\$85 = \$10.20$. Total = $\$85 + \$10.20 = \$95.20$.

20. **30%**
    Working: Profit = $\$156 - \$120 = \$36$. $(\frac{36}{120}) \times 100\% = 30\%$.

21. **$594**
    Working: After 10% increase = $\$660$. After 10% decrease = $\$660 - \$66 = \$594$.

22. **$560**
    Working: Annual interest = 4% of $\$500 = \$20$. $Total interest for 3 years = \$60$. $Total = \$500 + \$60 = \$560$.

23. **School A**
    Working: School A = $\frac{36}{45} = 80\%$. School B = $\frac{48}{64} = 75\%$. $80\% > 75\%$.

24. **70**
    Working: Other sports percentage = $100\% - (38\% + 27\%) = 35\%$. 35% of $200 = 70$.

25. **55%**
    Working: Shaded = $28 + 17 = 45$. Unshaded = $100 - 45 = 55$ out of $100 = 55\%$.

26. **70**
    Working: Coffee preference = 36%. Difference = $64\% - 36\% = 28\%$. 28% of $250 = 70$.

27. **25%**
    Working: Multiples of 4 are 4 and 8 (2 favorable outcomes). $\frac{2}{8} = \frac{1}{4} = 25\%$.

28. **78%**
    Working: Total counters = 50. Not blue = $14 + 25 = 39$. $\frac{39}{50} = \frac{78}{100} = 78\%$.

29. **15%**
    Working: Rectangle area = $12 \times 5 = 60$ cm^2. Cut-out square area = $3 \times 3 = 9$ cm^2. $(\frac{9}{60}) \times 100\% = 15\%$.

30. **44%**
    Working: Original area = 100 cm^2. New side = 12 cm, new area = 144 cm^2. Increase = 44 cm^2 out of 100 cm$^2 = 44\%$.